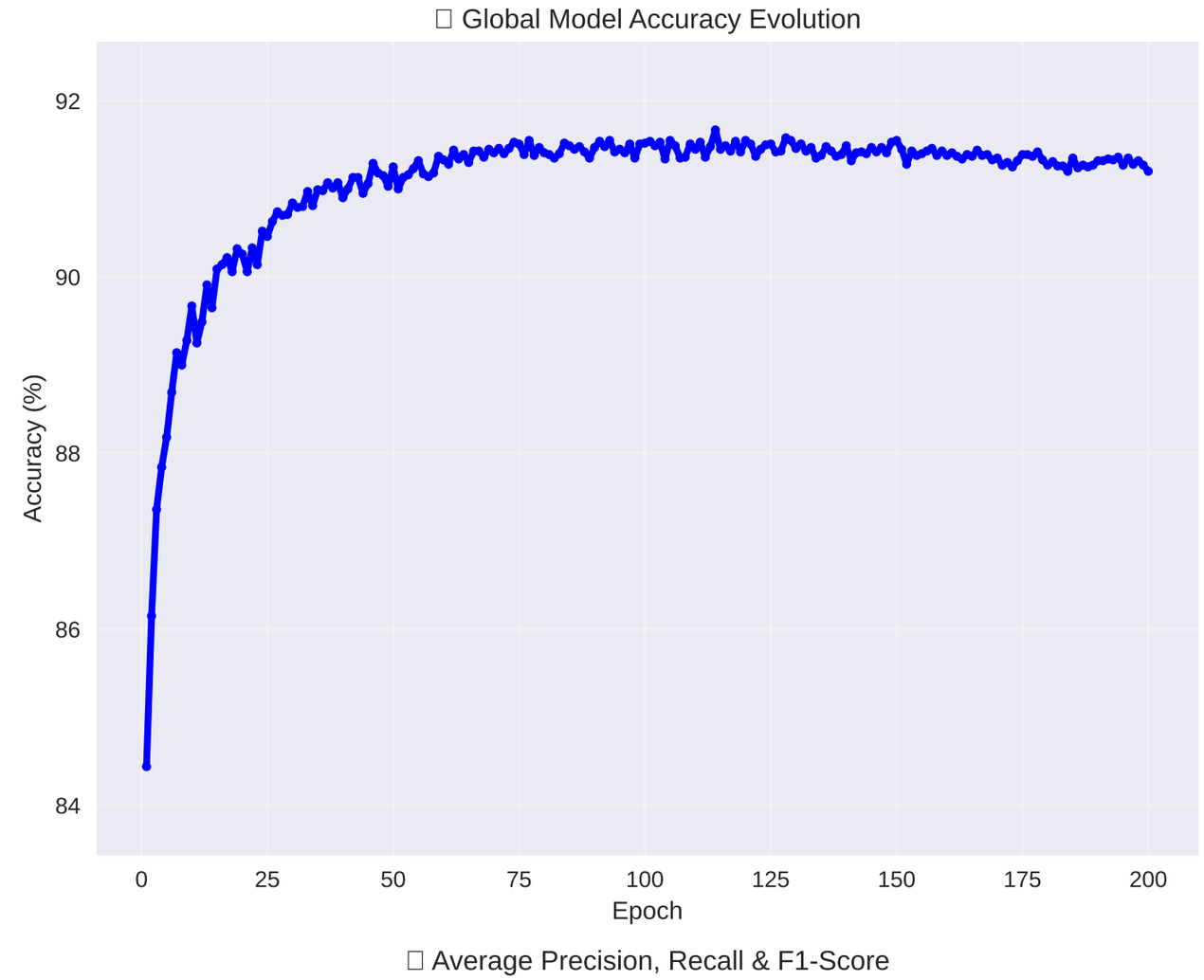
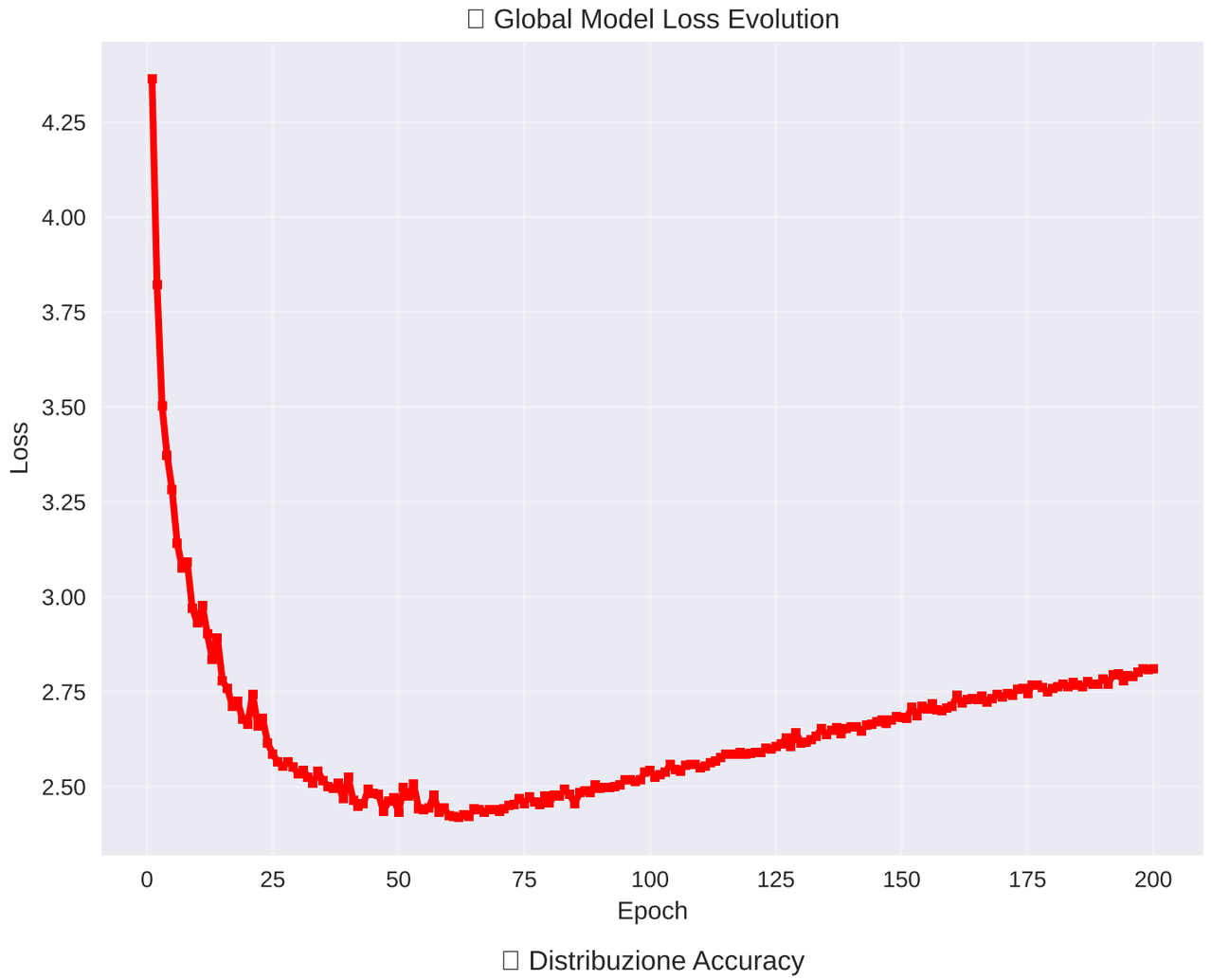
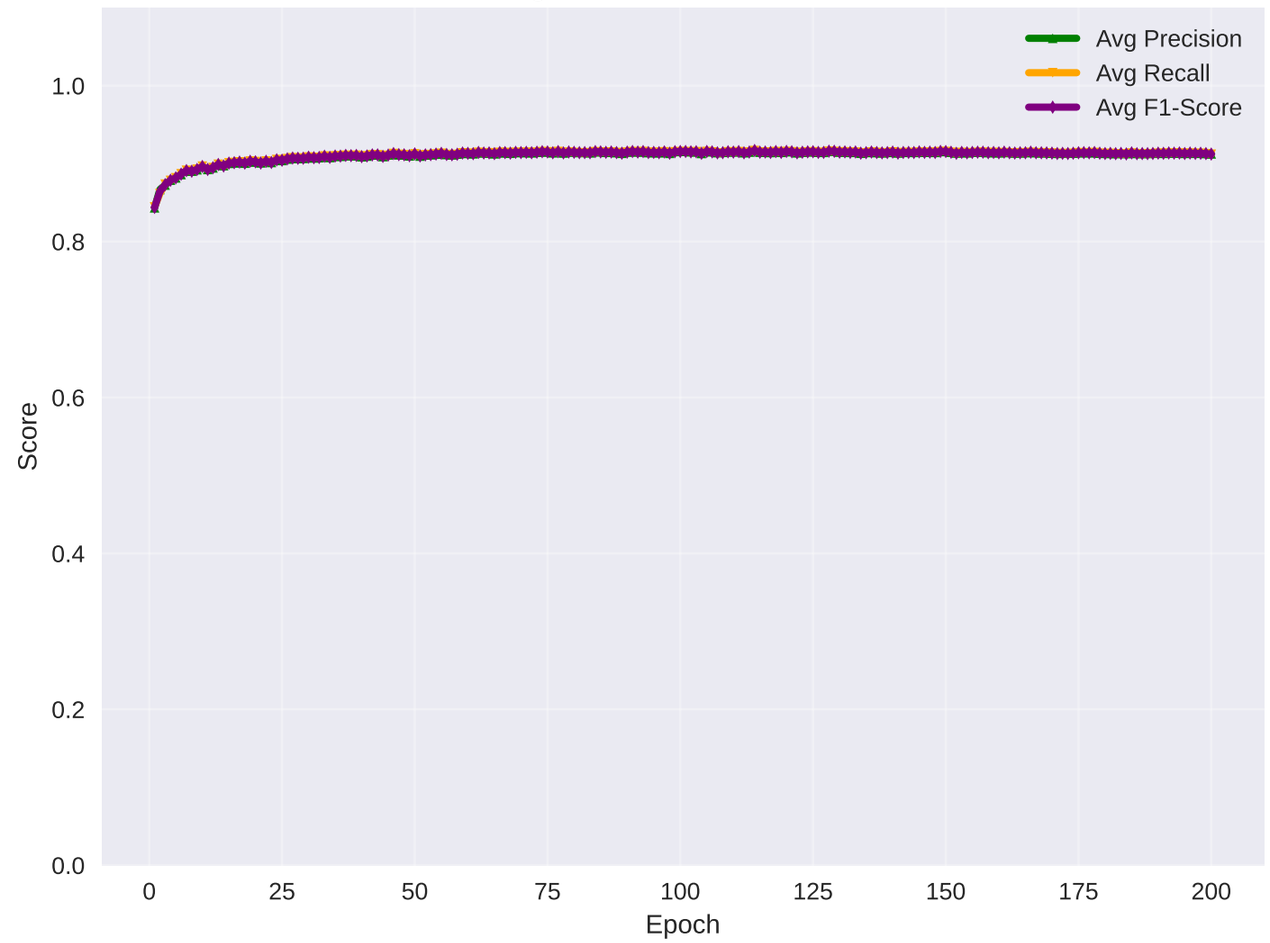
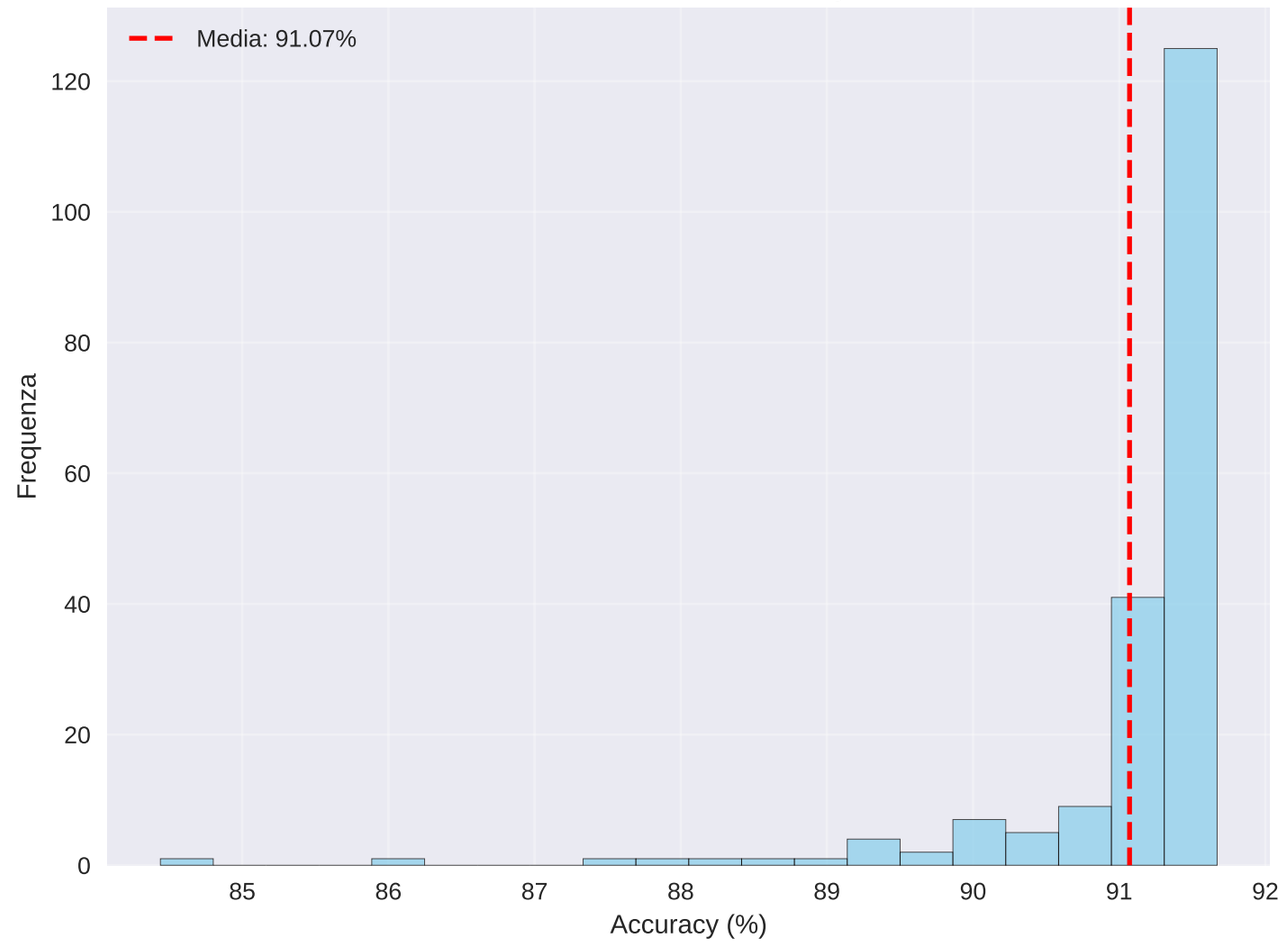
Federated Learning - Main Metrics Overview



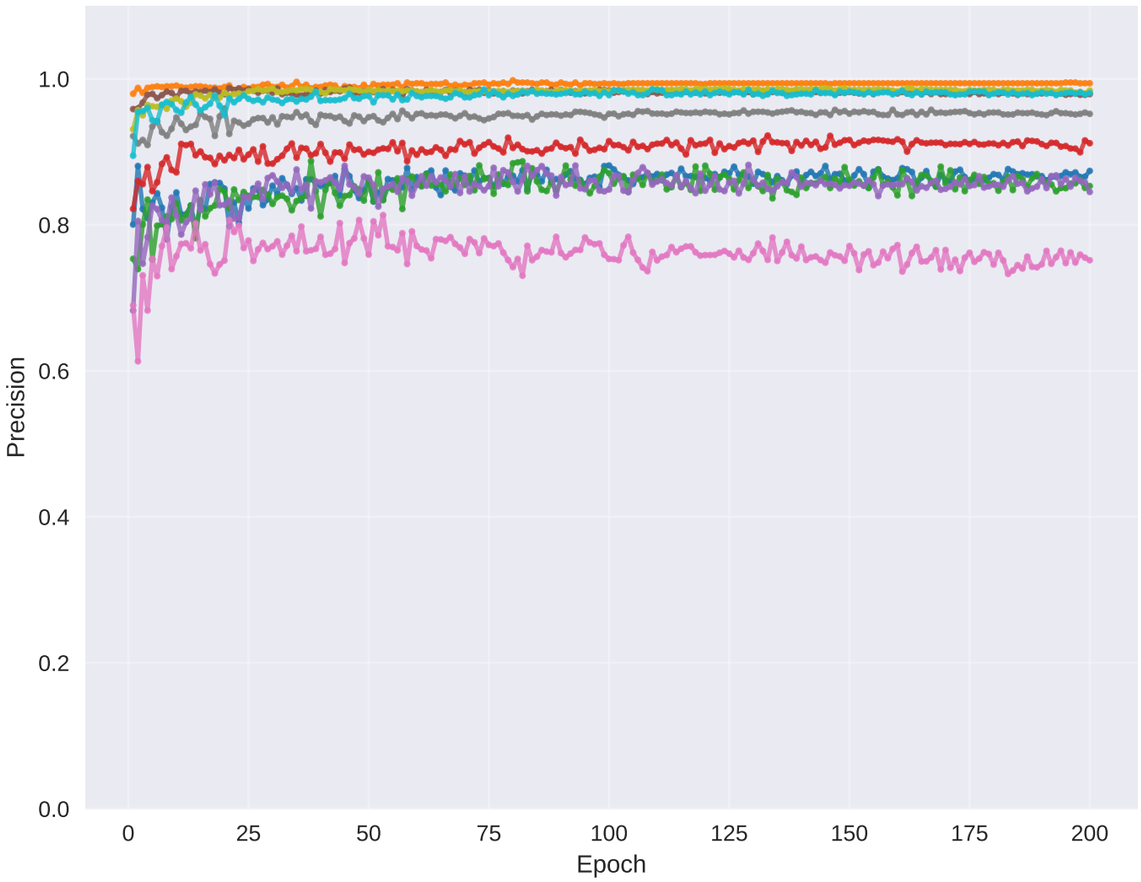




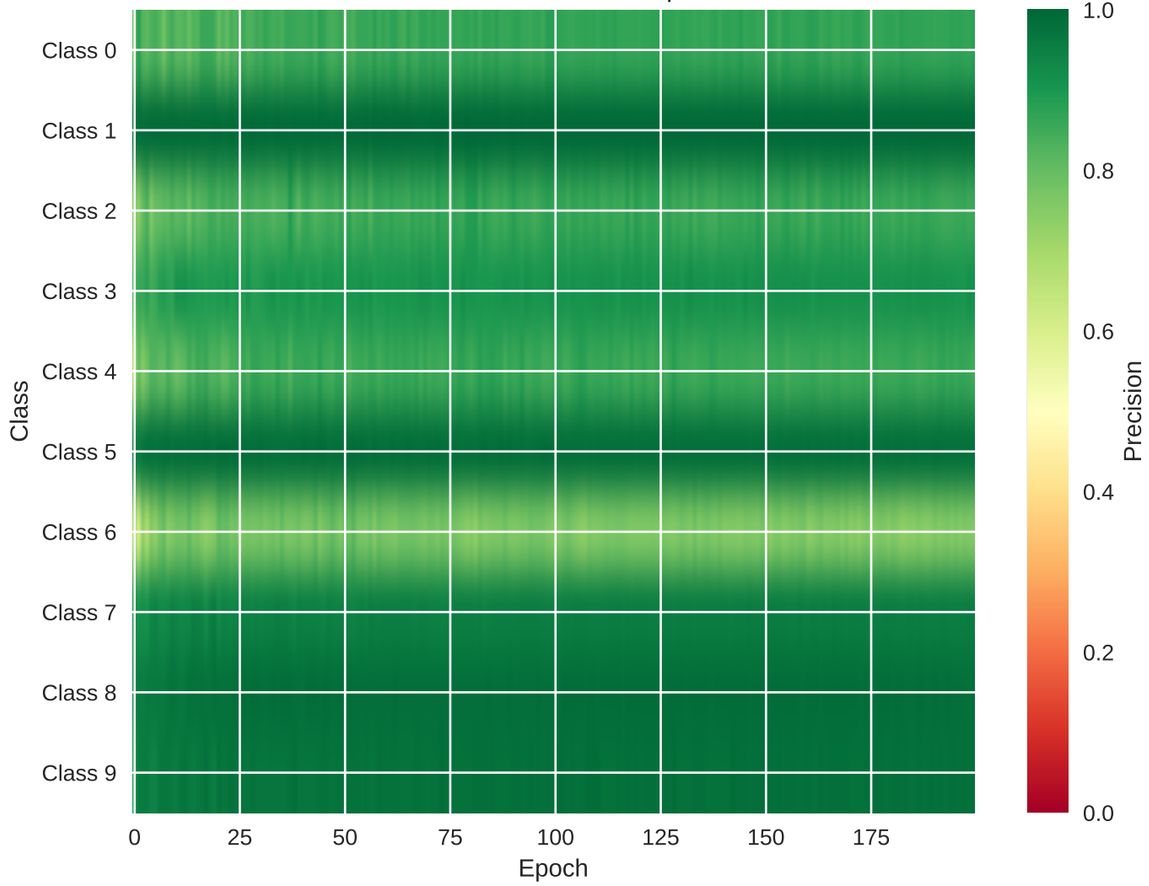


Per-Class Precision Analysis

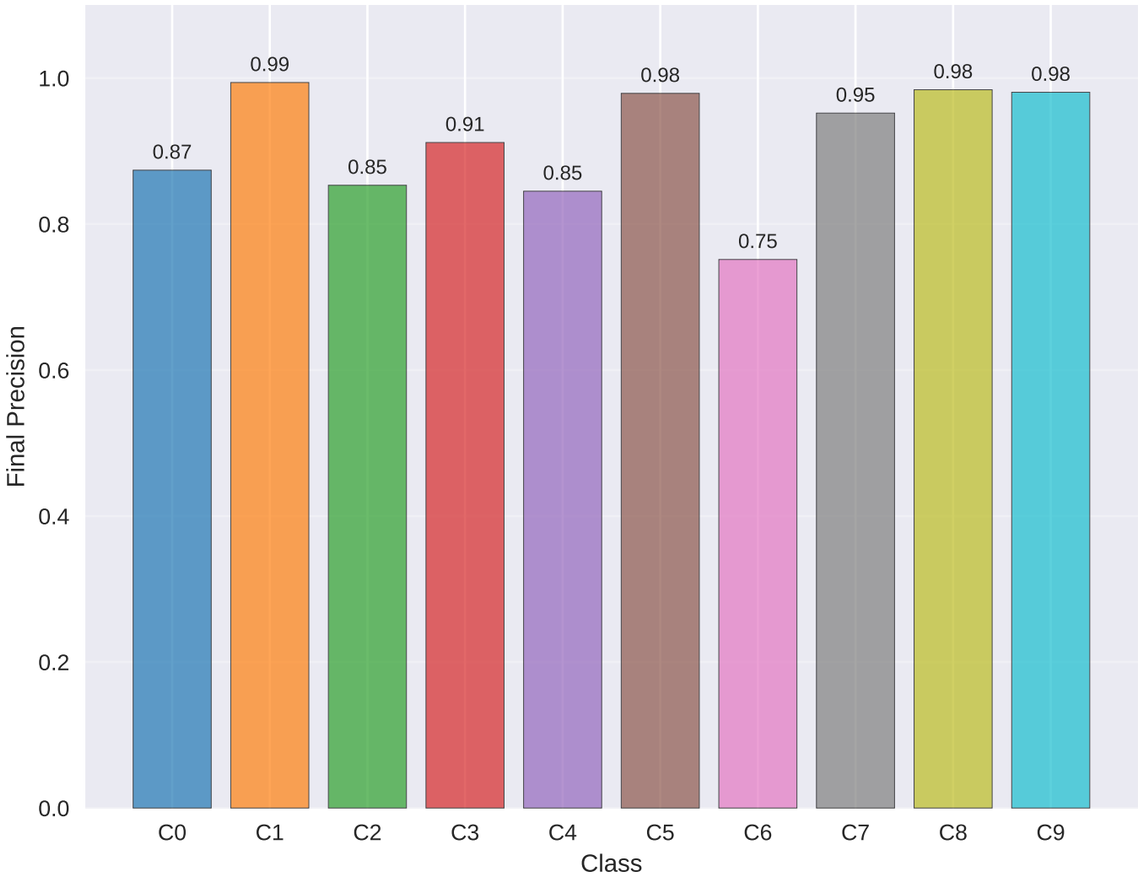
□ Per-Class Precision Evolution



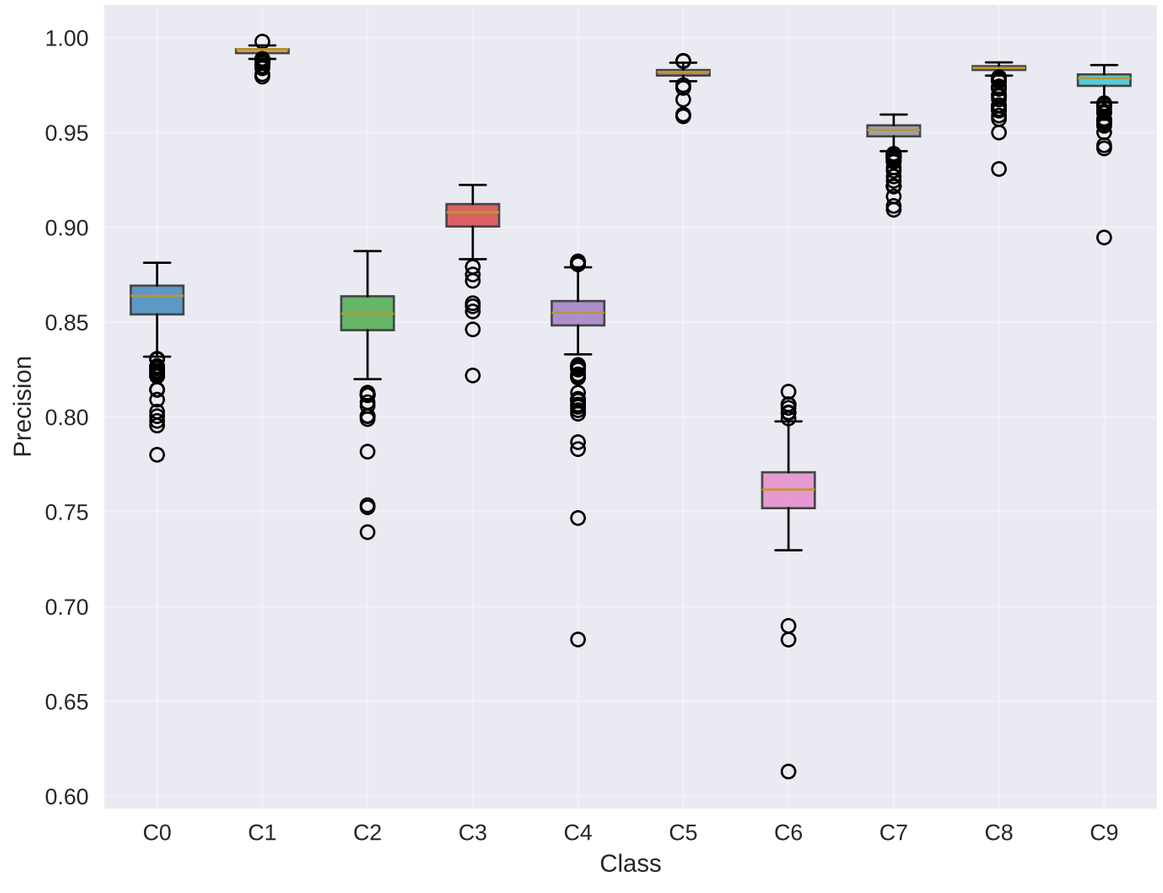
□ Precision Heatmap



□ Final Precision per Class

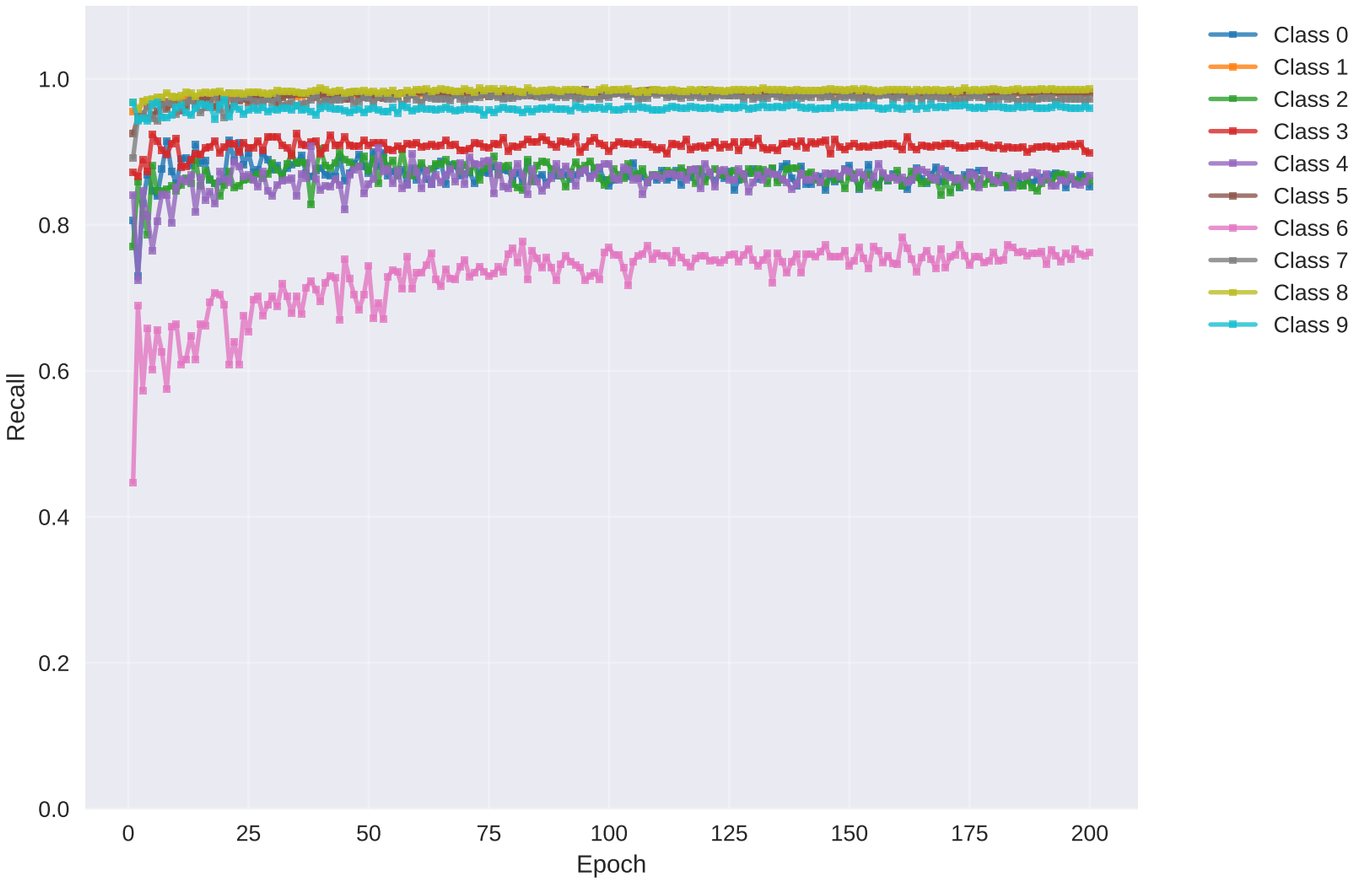


□ Precision Distribution per Class

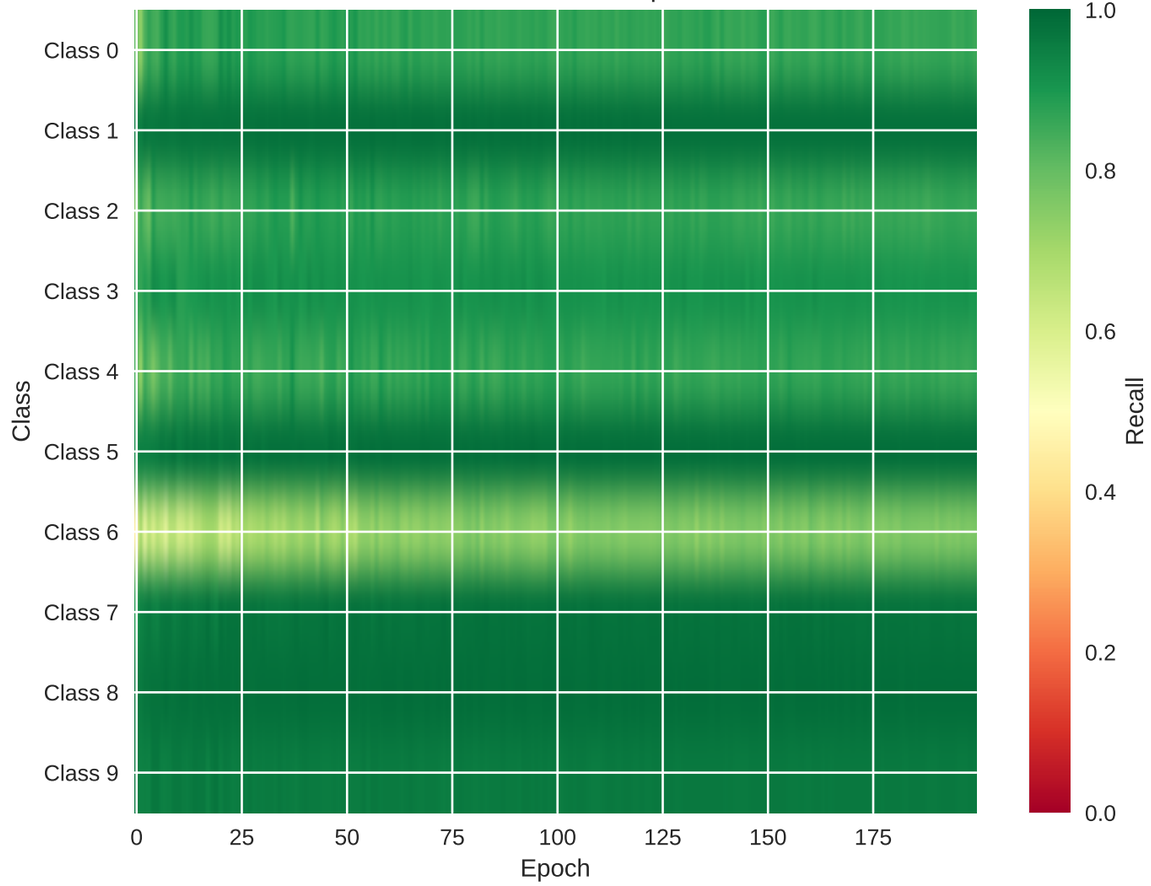


Per-Class Recall Analysis

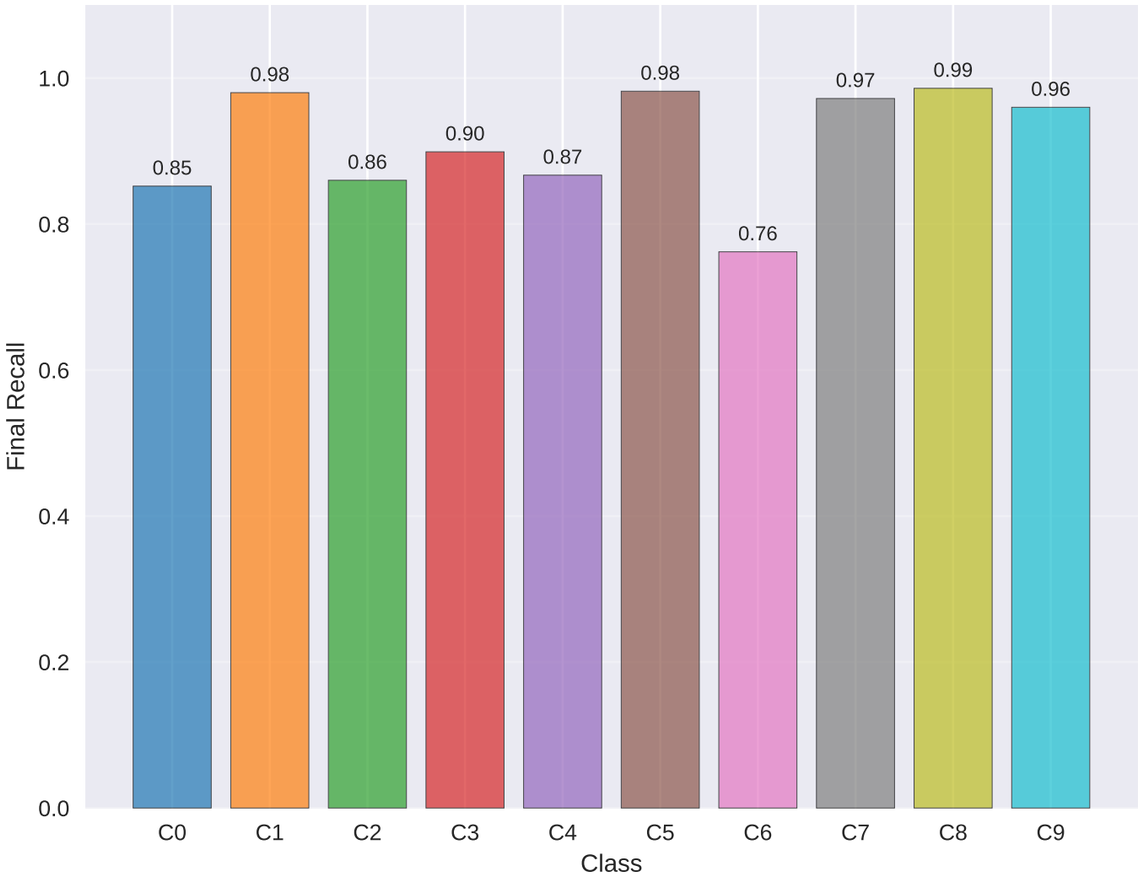
📈 Per-Class Recall Evolution



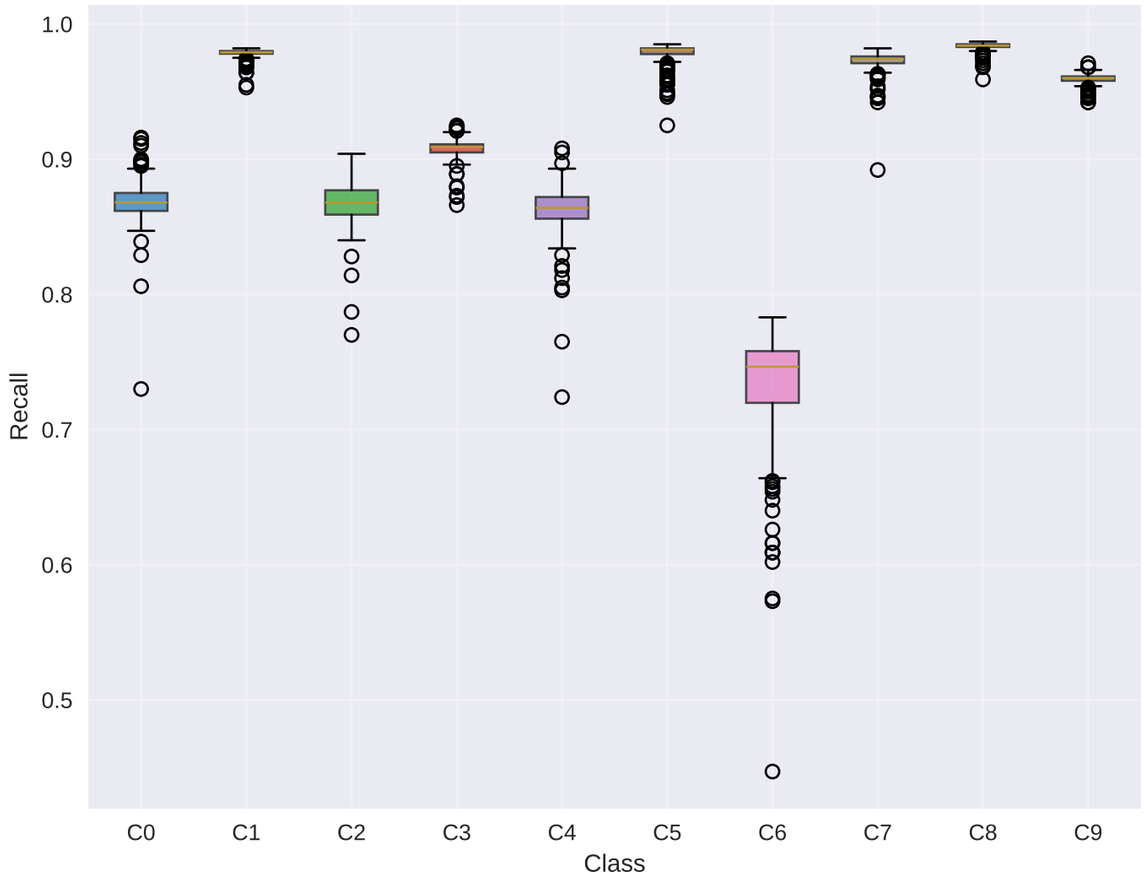
🔥 Recall Heatmap



🎯 Final Recall per Class

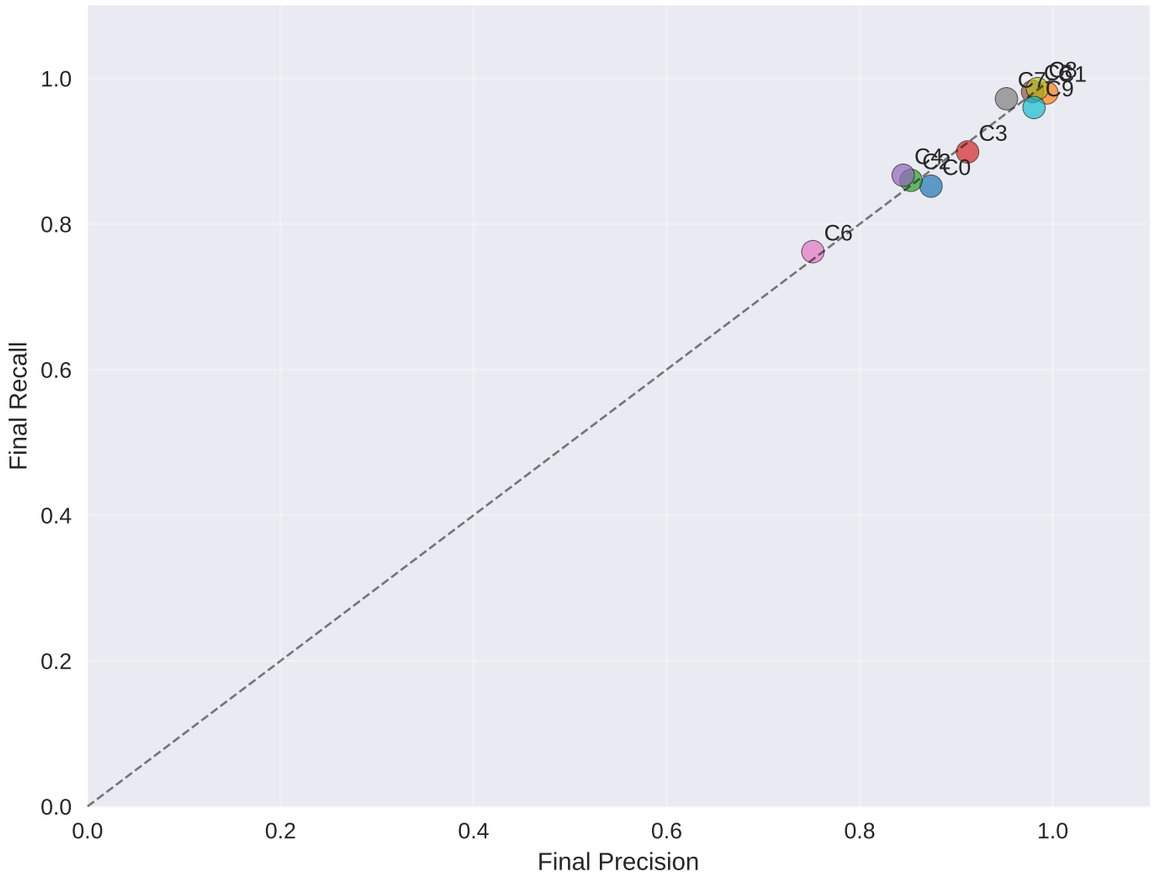


📦 Recall Distribution per Class

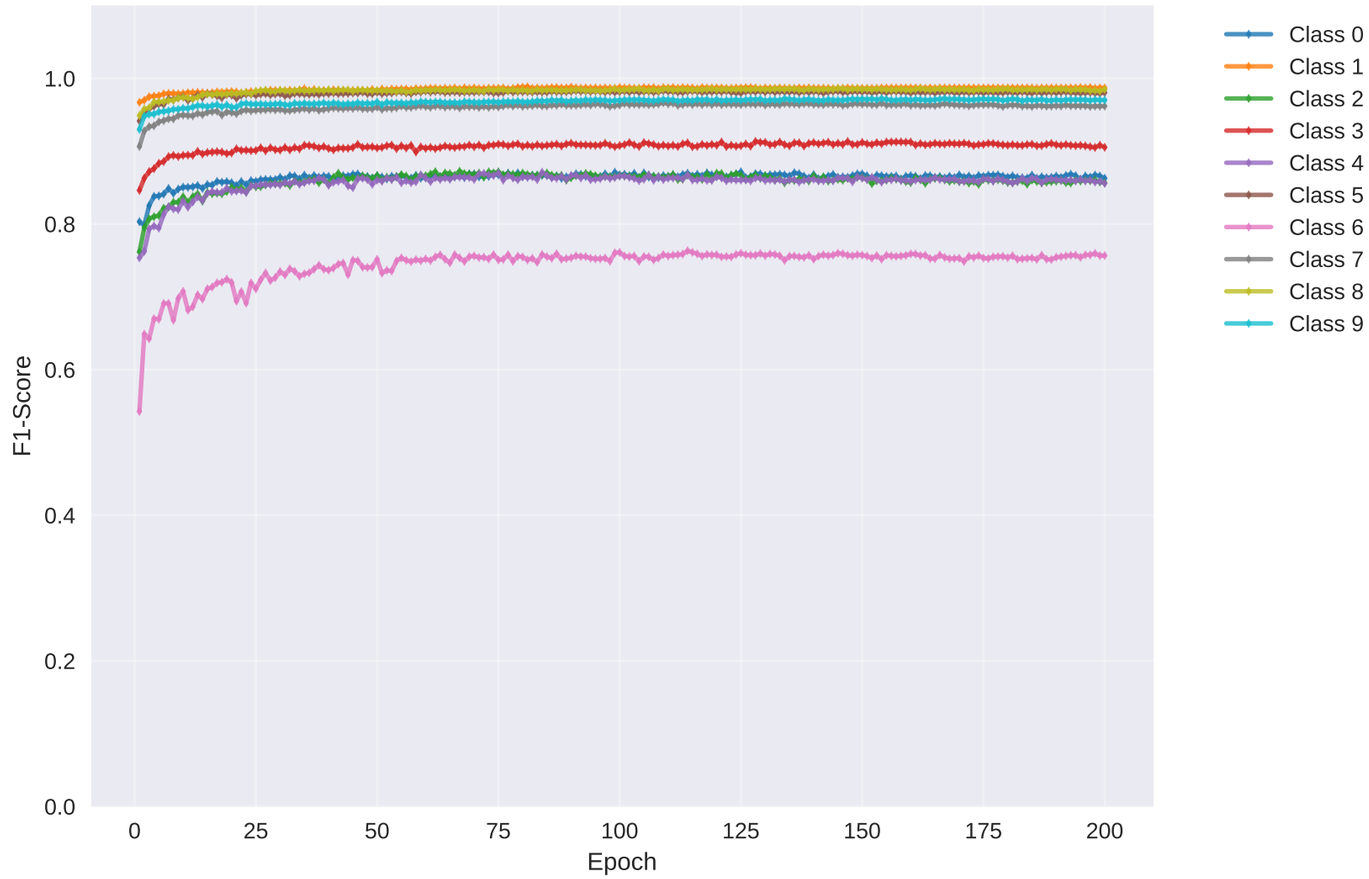


Comparative Analysis

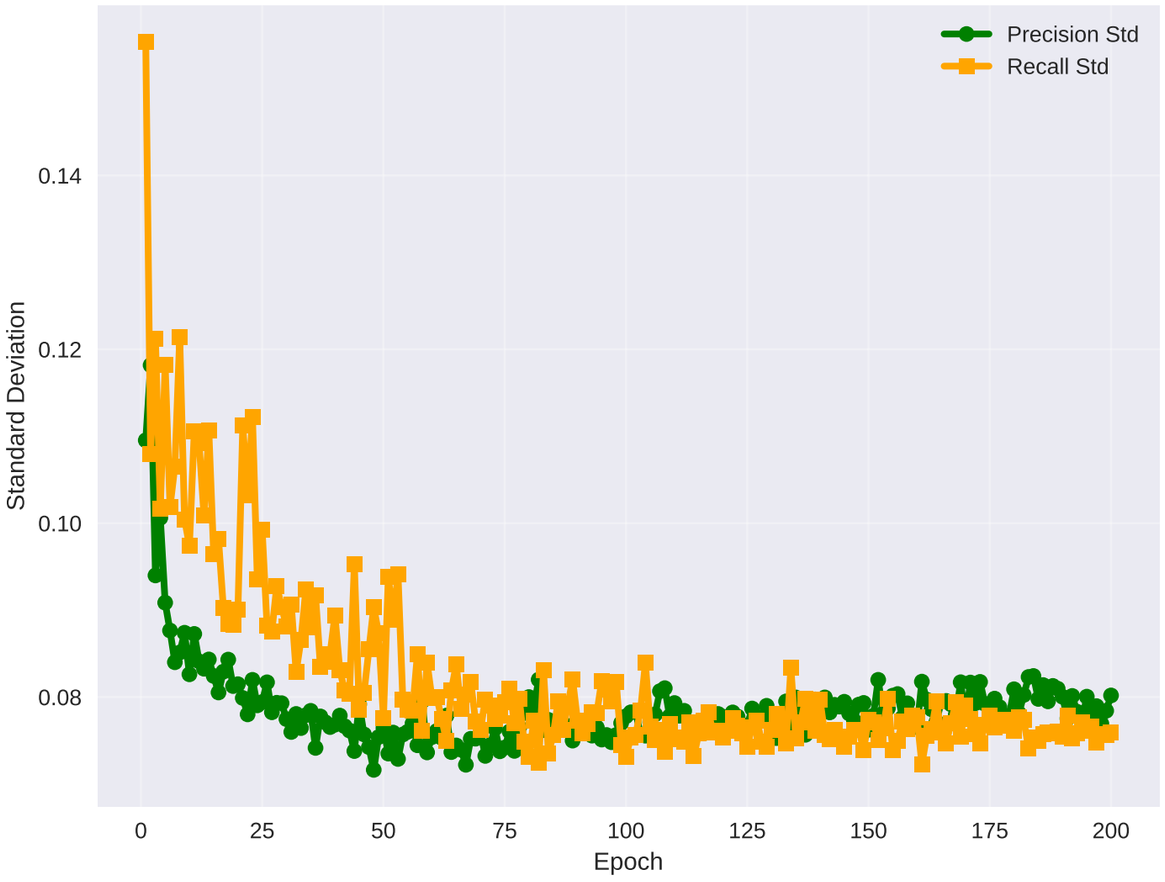
□ Precision vs Recall (Final Values)



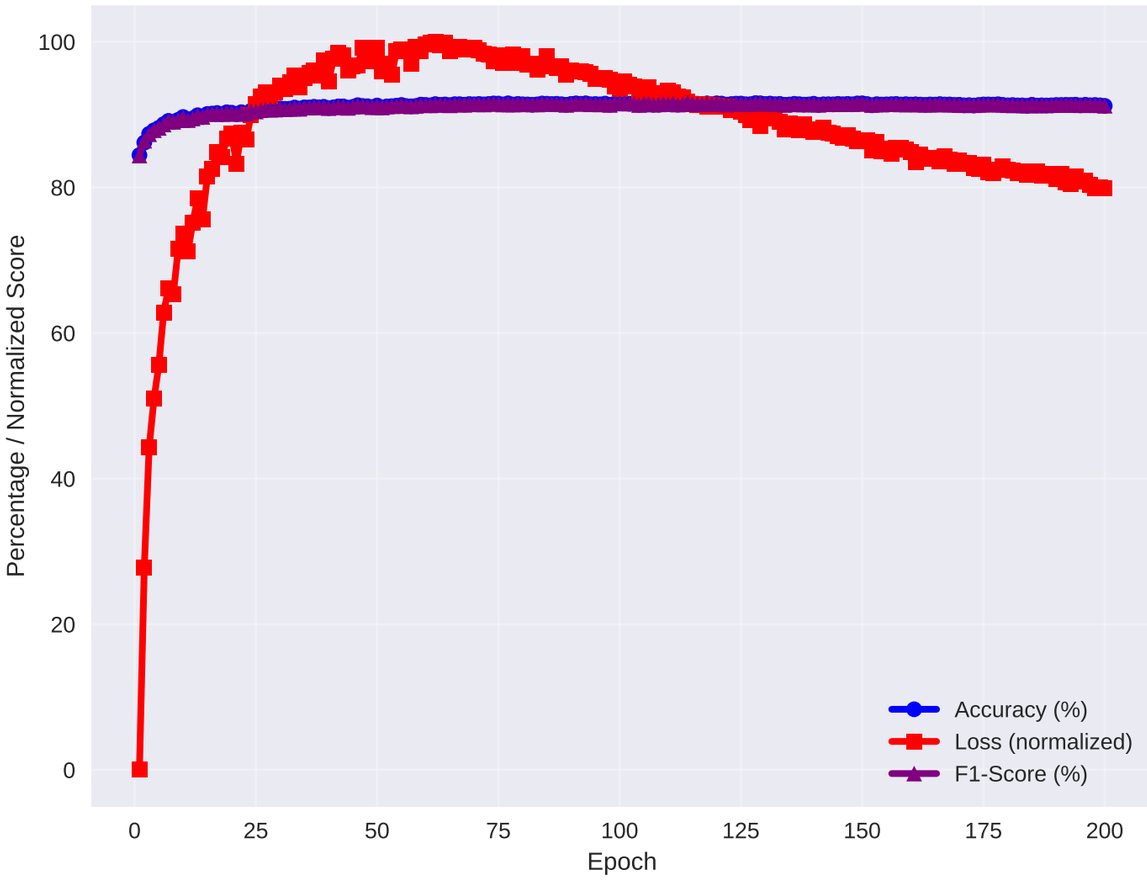
□ Per-Class F1-Score Evolution



□ Inter-Class Variance Evolution



□ Learning Curves Comparison



Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-02 19:03:12
EPOCHE TOTALI: 200

METRICHE FINALI:
• Accuracy: 91.20%
• Loss: 2.8107
• Average Precision: 0.912
• Average Recall: 0.912
• Average F1-Score: 0.912

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.76%
• Riduzione Loss: +1.5533
• Stabilità Accuracy (CV): 0.0097

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.994)
• Peggiore Precision: Classe 6 (0.751)
• Migliore Recall: Classe 8 (0.986)
• Peggiore Recall: Classe 6 (0.762)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.874, R=0.852, F1=0.863
• Classe 1: P=0.994, R=0.980, F1=0.987
• Classe 2: P=0.853, R=0.860, F1=0.857
• Classe 3: P=0.912, R=0.899, F1=0.905
• Classe 4: P=0.845, R=0.867, F1=0.856
• Classe 5: P=0.979, R=0.982, F1=0.981
• Classe 6: P=0.751, R=0.762, F1=0.757
• Classe 7: P=0.952, R=0.972, F1=0.962
• Classe 8: P=0.984, R=0.986, F1=0.985
• Classe 9: P=0.981, R=0.960, F1=0.970

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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